

Conjugate heat transfer for single phase immersion cooling of CPU

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ABSTRACT

This study develops a new design of a single-phase immersion cooling system for CPU in computers. The system consists of a loop filled with a dielectric liquid. Recirculation, which facilitates conjugate heat transfer, is driven by thermal convection without the assistance of a fan or pump. A key innovation which allows a sustained convective flow is the asymmetric geometry of the pipe, including a section of a multi-staged Tesla valve which restricts the fluid flow to one direction. The feasibility of the system is tested by CFD (computational fluid dynamics) simulations with two choices of liquid: mineral oil and FC-3283. It is found that the case with FC-3283 attains a greater degree of cooling compared to mineral oil. In both cases, the temperature of CPU is shown to be in the range that is suitable for real applications. Suggestions are made for future explorations of the geometric effect in closed-loop cooling systems.

1. Introduction

The Central Processing Unit (CPU) is the core of a computer where all heavy computations take place. Progresses in the past decades have led to increasingly powerful processors which also require an increased level of power consumption. Along with it, management of cooling for CPUs becomes an important challenge in engineering. A classical method of CPU cooling is by forced air flow driven by fans. While it is relatively easy to implement, the approach is limited by the material property of air, and reliance on extra mechanical parts in the fan (e.g., Refs. [1,2]). Immersion cooling by liquid or liquid-gas hybrid has been considered as an alternative. The availability of a wide selection of liquids with different thermal properties allow potential applications to high-power systems (e.g., Refs. [3–5]). In the past decade, immersion cooling has undergone significant development, for systems ranging from a single server (e.g. Refs. [6,7]) to large arrays for a data center (e.g., Ref. [8]). A prominent design is to use a closed recirculating loop to facilitate conjugate heat transfer. This broadly belongs to the class of thermosyphon, e.g., see the surveys of [9,10] and further developments in Refs. [9–12]. When combined with phase-change mechanisms, various designs of two-phase thermosyphons have shown promise for cooling of high-power systems ([12–14]).

In general, efficient conjugate heat transfer in the closed loop requires the flow to be predominantly uni-directional. This flow can be maintained by phase-change mechanisms, or by mechanical drive with a fan or pump. As a less explored aspect of it, asymmetry in the geometry of the loop may also help enhance the uni-directional behavior of the convective flow. Given this background, the goal of this study is to explore this aspect in a relatively simple setting. We take a barebone design of a closed loop with single-phase liquid and without an internal fan. The device is small enough to fit within a stand-alone workstation, e.g., a mini tower equivalent to a 2U server. Based on it, the geometry of the pipe is modified by the insertion of a strongly asymmetric segment, namely, a multi-staged Tesla valve with a high diodicity for fluid flow. By computational fluid dynamics (CFD) simulations, we demonstrate that the design permits a strongly uni-directional convective flow that facilitates cooling to the satisfactory range for real applications, for a chip with a modest

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power output.

Our contributions are twofold. First, by demonstrating the aforementioned working design, we draw attention to the fact that using highly asymmetric geometry for the loop could potentially improve existing designs of a general thermosyphon. Of course, the effect of asymmetric geometry alone does not supersede other existing and important mechanisms (phase change, mechanical drive, etc.) Rather, we view it as a potential add-on feature to enhance those existing designs. Secondly, we also cast this study as a step forward in expanding the applications of Tesla valve (surveyed in Section 2). Despite a recent revival of interests in the device, little has been done to put Tesla valve in the context of closed-loop, buoyancy-driven flow. This study bridges the gap, and opens up new opportunities to explore the intersection of researches on fluid diodicity and conjugate heat transfer.

2. Technical preliminaries and preparation

2.1. The recirculation system

A prototype of the closed-loop cooling system is sketched in Fig. 1a. It consists of a recirculating 3D pipe filled with a single-phase liquid. The loop is oriented with the hot CPU placed at bottom, and a cooling zone at top. Using an asymmetric geometry of the pipe to restrict the fluid flow to predominantly one direction, a flow can be maintained by thermal convection due to buoyancy effect. The circulating flow facilitates conjugate heat transfer. A naive design with weak asymmetry such as shown in Fig. 1a is usually not enough to restrict an uni-directional flow, and a higher degree of asymmetry of our actual design will be unveiled in Section 3. Before proceeding, we also note that the component marked as CPU in Fig. 1a consists of a thin plate of the computer chip at the center, surrounded by a heat spreader. This detail is shown separately in Fig. 1b. The CPU and heat spreader are shown in red and gray, respectively. They together are further attached to a fin-like heat sink (yellow in the top panel of Fig. 1b), which is immersed in the liquid and allows the flow to pass through the gaps between the fins. While Fig. 1a is generic, Fig. 1b shows the actual geometry of the CPU component to be used in our full design.

2.2. Selection of cooling medium

We next choose the liquid in the pipe. Surveying previous studies ([3–8]), Mineral Oil (e.g., Silicone Oil 20, Silicone Oil 50) and 3M™ Fluorinert (e.g., FC-43, FC-3283) are the most commonly used for electronic cooling. This is mainly due to their desirable thermal properties (thermal conductivity, specific heat, etc.). To restrict our design to single-phase cooling, the liquid should have a sufficiently high critical temperature for evaporation. Putting together these considerations, we select two liquids, a generic Mineral Oil and 3M™ Fluorinert FC-3283, to test our design. Their key properties are listed in Table 1, using information from Refs. [15,16] for Mineral Oil and [17] for FC-3283. Note that Mineral Oil and FC-3283 have much higher dielectric strengths (11.8 and 40 kV/mm, respectively) compared to air (3 kV/mm). The density (ρ) and thermal expansion coefficient (β) listed in the table are those at the reference temperature of $T = 308\text{K}$. By Boussinesq method (to be used in CFD simulations), the thermal expansion coefficient is given as $\beta = -\rho^{-1}(d\rho/dT)$, computed at the reference temperature.

3. Key design and CFD assessment

3.1. Application of Tesla valve

In designing the recirculating system, we first tried simple geometry with weak asymmetry, similar to that shown in Fig. 1a. It is found that such designs do not work, as a strong unidirectional flow never forms. After a series of trials (some details are in Ref. [18]), a significant improvement is attained with strongly asymmetric geometry. The final design is shown in Fig. 2a. The key insight is in replacing the bottom section of the pipe by a tilted multi-staged Tesla valve.

Tesla valve was invented by Nikola Tesla in 1916 [19]. Fig. 2b, after Tesla's original sketch, illustrates how it facilitates flow control. Given the same pressure drop, a strong forward flow (from A to B) can develop without impediment, while a reverse flow (from

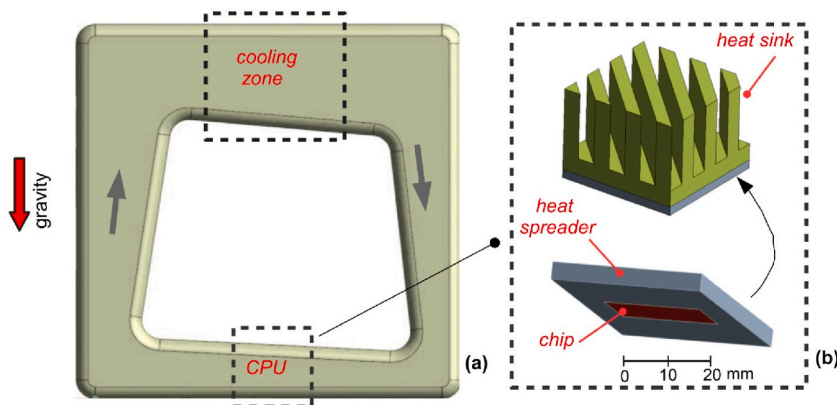


Fig. 1. (a) A sketch of a prototype of the recirculating cooling system with weak asymmetry in geometry. (b) Detail of the CPU component, which is used in the real design in Section 3.

Table 1
Properties of the two liquids used as cooling medium.

Properties	Mineral Oil	FC-3283
Density [kg/m^3]	856.04	1816
Specific Heat [$\text{J}/\text{kg} \cdot \text{K}$]	1917.83	1046.63
Dynamic Viscosity [$\text{N} \cdot \text{m}^2/\text{s}$]	0.02340	0.0014
Thermal Conductivity [$\text{W}/\text{m} \cdot \text{K}$]	0.1488	0.063824
Thermal Expansion Coefficient [$1/\text{K}$]	7.3475×10^{-4}	0.0014
Thermal Diffusivity [m^2/s]	9.0854×10^{-8}	3.404×10^{-8}
Dielectric Strength [kV/mm]	11.8	40

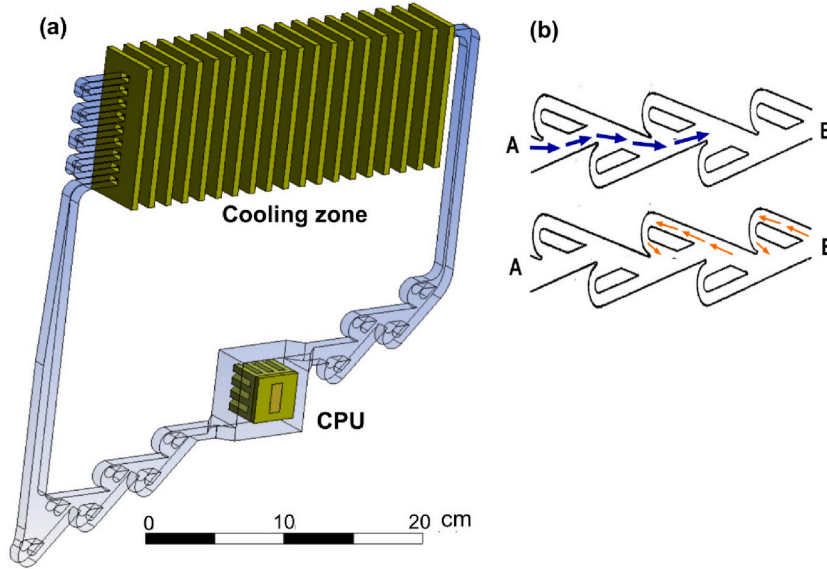


Fig. 2. (a) The design of the cooling system. The fluid domain is in blue, and CPU and radiator in the cooling zone are in brown. (b) A sketch of the asymmetric property of the Tesla valve. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

B to A) is weak because it is blocked. To achieve the same flow rate, the pressure drop for the reverse flow (ΔP_r) would have to be greater than that for the forward flow (ΔP_f). Their ratio, the *diodicity* $D = \Delta P_r / \Delta P_f$, quantifies this asymmetric flow property ([20,21]). A desirable design will have $D \gg 1$. Tesla valve has seen applications in flow control for various fluid systems (e.g. Ref. [22]). Recently, it began to be used in heat transfer problems (e.g., Refs. [23,24]). The study in Ref. [25] uses Tesla valve to control heat transfer in a recirculating loop, although for a system that is significantly different from ours.

The whole cooling system has the dimension of less than $40 \text{ cm} \times 45 \text{ cm} \times 6 \text{ cm}$ (see the scale in Fig. 2a), small enough to fit in a typical desktop computer. The CPU unit is placed in the middle of the multi-staged Tesla valve, which is tilted at a 45° angle. The cooling zone is placed at the top. With hotter (and lighter) fluid in the CPU box and cooler fluid in the cooling zone, the Tesla valve helps guide the buoyancy-driven flow in the counterclockwise direction as viewed in Fig. 2a. The two vertical segments of the pipe are relatively thin, with narrowest cross-sectional area of approximately $5 \text{ mm} \times 10 \text{ mm}$. This is to prevent the formation of (vertically oriented) vortices, which would decrease the strength of the recirculating flow.

3.2. CFD assessment

To assess the performance of the design, we conduct CFD simulations for the system using ANSYS-Fluent (ansys.com). The geometry is as shown in Fig. 2a. Two choices of liquid, Mineral Oil and FC-3283, are tested. As noted before, Boussinesq method is used to treat thermal convection. Using 308K as the reference temperature, the thermal expansion coefficient and other relevant parameters are listed in Table 1. The CFD solver in ANSYS-Fluent employs the full set of Navier-Stokes (momentum) equations, continuity equation, and thermodynamic energy equation. The effects of both thermal convection and conduction are included. For thermal convection, by Boussinesq approximation the buoyancy term in the vertical component of momentum equation is represented by $\beta(T - T_0)g$, where the thermal expansion coefficient (β) and reference temperature (T_0) are as described in Section 2.2 and Table 1, and g is gravity. Steady solution is sought by iterations until the normalized residuals are sufficiently small. We use the comprehensive CFD solver as the system has complicated flow structures coupled to heat transfer. A simplified analytic calculation would not account for all the details.

The detail of the CPU component is as shown in Fig. 1b. The chip at the center (red in Fig. 1b) has the dimension of $22.4 \text{ mm} \times 9.2 \text{ mm}$ and thickness of 0.58 mm. The heat spreader (gray in Fig. 1b) is $35 \text{ mm} \times 35 \text{ mm}$, with a thickness of 2.59 mm. The chip is made of

silicon; Heat spreader and heat sink (yellow in Fig. 1b, which is submerged in the liquid) are made of copper. The heat sink has carved channels oriented at 45° angle to align with the direction of the prevailing flow along the Tesla valve. For the benchmark simulation (shown in Figs. 3–5), the power density of the chip is set to 112 W/cm^2 , equivalent to 241 W of power output. These settings mimic a commonly used chip, Intel Core i9-10900K, as detailed in Ref. [26]. The cooling zone at the top of the system consists of radiator panels (yellow in Fig. 2a) which serve to cool down the liquid. Except for the chip and radiator, the outer boundary of the fluid domain is treated as thermally insulated.

With the setting described above, Fig. 3 shows the temperature on the central plane from the steady solution, using FC-3283 as the cooling medium. Fig. 4a is a snapshot of the temperature at the outer boundary of the system. Note that the two figures have different temperature ranges and are presented with different color scales accordingly. Fig. 3 is for a cross section that cuts through the solid part of the heat sink, which has a significantly higher temperature than the fluid part. Fig. 4a is the temperature for fluid only (except the radiator), at the outer boundary of the opposite side of where the chip core is located. The detail of the temperature at the surface of the CPU component is shown in Fig. 4b.

The result demonstrates the behavior of hot liquid rising from the vicinity of the CPU, driving the flow into the cooling zone. The cooled liquid then circulates back to the CPU box. The continued recirculation sustains conjugate heat transfer, and the CPU reaches a sufficiently low equilibrium temperature of 320.1K (46.9C).

The structure of the velocity field on the central plane from the same CFD simulation is shown in Fig. 5a. The contour plot is for the x-velocity; positive means the flow is from left to right. The prevailing flow in the Tesla valve is right to left and ascending. To further confirm this impression, the velocity vectors in a zoomed-up portion of the Tesla valve is shown in Fig. 5b. The maximum velocity in the section is approximately 3.5 cm/s.

With encouraging results from the benchmark run, we perform 4 additional runs that retain all conditions but with the power density of the chip changed to 90, 100, 105, and 120 W/cm^2 . These five cases are then repeated but with FC-3283 replaced by Mineral Oil. The equilibrium temperature of the CPU from the 10 runs are summarized in Table 2. The FC-3283 cases consistently yield a lower temperature, compared to their counterparts in Mineral Oil cases. For the benchmark case with power density of 112 W/cm^2 , the CPU temperature for the Mineral Oil case is 329.9K (56.7C), compared to 320.1K (46.9C) for the FC-3283 case. In both cases, the chip temperature falls within the satisfactory range for real applications. (The typical CPU temperature from published reports range from 45 to 55C for smaller servers (e.g., Ref. [27]) to 75–85C for larger systems (e.g., Ref. [28]) using commercial chips of a comparable class as ours, and with conventional air cooling. Similar ranges are reported at industrial websites, e.g., hwbusters.com/cpu). In our case, this is achieved without using a fan to drive the recirculation. Lastly, we note that 3M™, the manufacturer of FC-3283, might discontinue production of the compound in the future. Even so, our design still shows potential if Mineral Oil is used as the cooling medium.

4. Discussion

Our work demonstrates that highly asymmetric geometry may help enhance the convective flow in a closed-loop cooling system. The geometric “valve” effect is but one of several mechanisms that can be used to sustain the uni-directional flow in the loop. As the Tesla-valve design is tested on a relatively simple system, we by no mean claim that the geometric effect alone supersedes other existing mechanisms (by mechanical drive, phase change, etc.) for driving the recirculation. Nevertheless, the asymmetric geometry can potentially be adapted as an add-on feature to existing systems with those other mechanisms to enhance their performances. Our

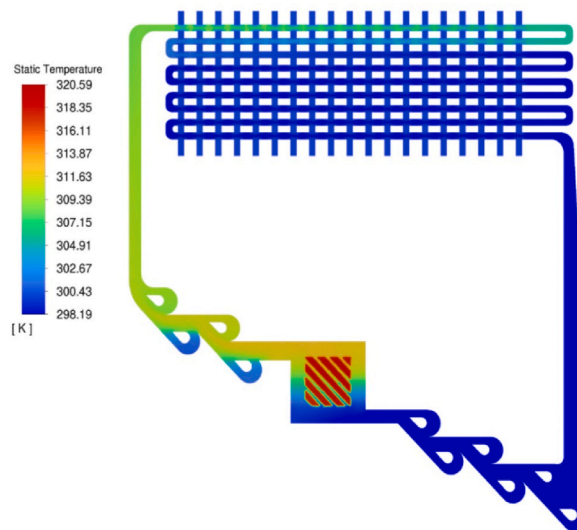


Fig. 3. Contour plot of temperature on the central plane of the cooling system. The case shown here and in Figs. 4–5 is the benchmark run using FC-3238 as cooling agent, and with power density set at 112 W/cm^2 and power output at 241 W for the chip.

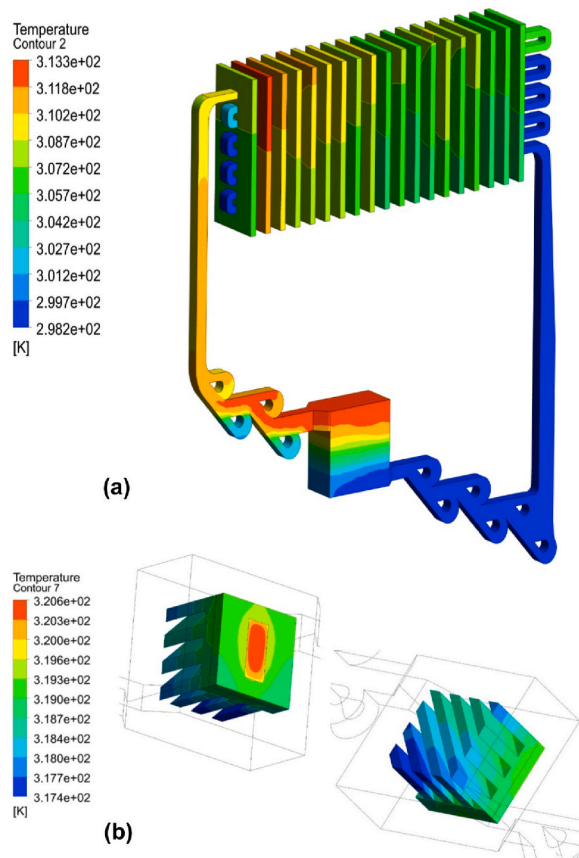


Fig. 4. (a) Temperature at the outer boundary of the system. (b) Two different views of temperature at the surface of the CPU unit, including the chip, heat spreader, and heat sink.

design is tested using a chip with a modest power output. In the high-power regime, extra mechanisms such as phase change would likely be more dominating in facilitating the needed cooling. Since we only consider a single-phase design, the high-power regime is not explored here. It is also relevant to note that the behavior of Tesla valve for multiphase flow is not yet well-understood (for example, the diodicity would be different for different phases and the mixture), which by itself is another interesting problem for future work.

Our current design could be further optimized. To do so systematically would require a better understanding of how the flow rate of the buoyancy-driven flow depends on key thermal and geometric parameters. This involves the interplay between the parameters on the thermal side, such as Rayleigh number, and those on the fluid side, particularly the diodicity of the Tesla valve. The system we tested on is relatively small in size. Whether it can be scaled up (and still maintain a similar level of performance) remains to be investigated. Previous studies on thermosyphon have explored various aspects of scaling. For example, it is shown in Ref. [9] that the height difference between the heating element (our chip) and the cooling zone is a relevant parameter for the flow in a thermosyphon. We should include such parameters in future studies.

Another aspect of future work is to optimize the fin-like heat sink, which in our design is embedded in the middle of the multi-staged Tesla valve. (It is not trivial that we orient the channels between the fins in the direction of the forward flow in the Tesla valve.) In that sense, the geometry of the heat sink and the multi-staged Tesla valve should be adjusted together. To prepare for it, a more detailed thermofluid analysis surrounding the heat sink and nearby Tesla valves (e.g., in the fashion of [29]) will be useful. Lastly, the assessment of our design relies on CFD simulations. Further testing of similar systems in the laboratory will be welcome.

5. Conclusion

This study achieves the proof of concept that incorporating highly asymmetric geometry in a recirculating loop helps sustain a convective flow for conjugate heat transfer. The idea is realized by an example of a small-scale system for CPU cooling, using a single-phase liquid as cooling agent. Numerical simulations by a CFD solver indicate that the system can bring CPU temperature to the normal range for real applications; The equilibrium temperature is comparable to reported ranges for smaller servers using a similar class of chip, and with conventional air cooling. Our system sustains this temperature range without the assistance of a fan.

The geometric effect, crucially established by a multi-staged Tesla valve, is tested on a relatively small system with a modest power

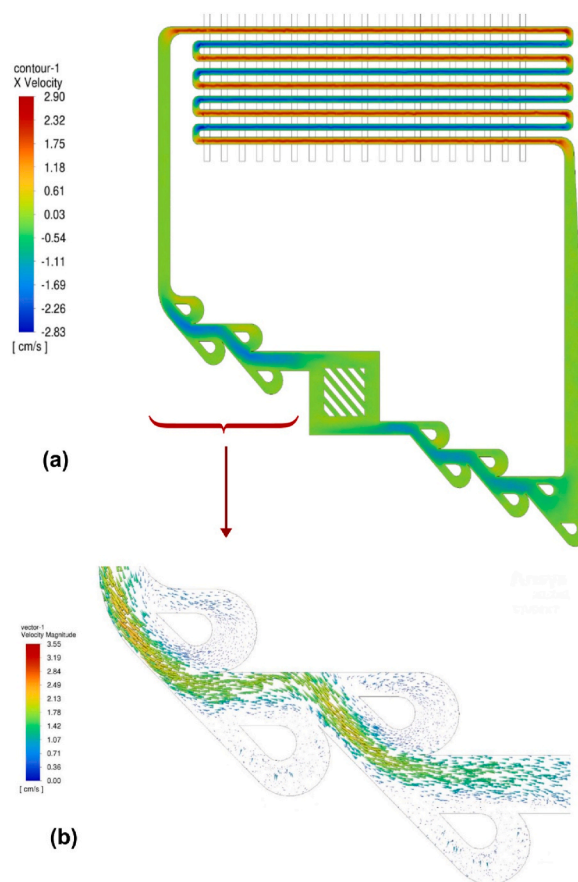


Fig. 5. (a) Contour plot of the x-velocity on the central plane. Red is positive (flow from left to right) and blue is negative. (b) Detail of the velocity vectors in a section (indicated in (a)) of the tesla valve. The color scale is for velocity magnitude. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Table 2

The equilibrium temperature of CPU from 10 CFD simulations with different imposed power density of the chip (left column). Center and right columns show the results for the FC-3283 and Mineral Oil cases, respectively.

Power density of chip [W/cm^2]	CPU temperature [K] FC-3283 case	CPU temperature [K] Mineral Oil case
90	316.8	322.9
100	318.4	326.4
105	319.2	327.9
112	320.1	329.9
120	321.2	331.9

output for the chip. Previous studies (as surveyed in Section 1) have demonstrated thermosyphons driven by other mechanisms such as phase change that work for a wide range of power. The Tesla valve mechanism is not meant to supersede those existing designs, but complement them as an extra aspect to consider to optimize the cooling system.

In the context of recent developments on high-diodicity fluid valves, our work opens up possibilities of incorporating Tesla valve in closed-loop buoyancy-driven flows, which until recently was rarely explored. Our result is yet another demonstration of the versatility of Tesla valve in new engineering applications.

For real-world impact, we envision that adaptation of asymmetric geometry in suitable classes of thermosyphon could help increase their performances in energy applications. Our current design of a smaller single-phase system, by itself, might serve a niche market for personal computing in a modest range of power. Our cooling system may allow the computer to operate more quietly while eliminating power consumption by the fan.

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Author statement

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Declaration of competing interest

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Data availability

No data was used for the research described in the article.

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